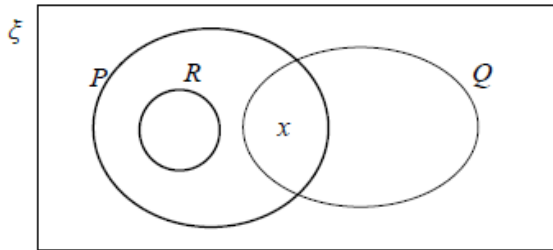
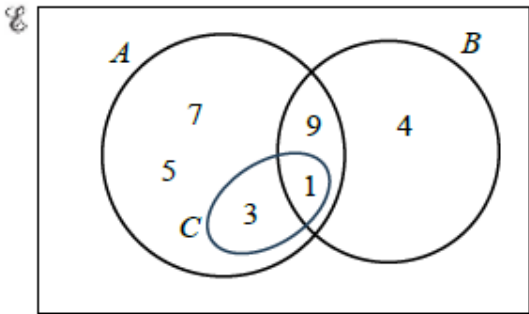


1	YTSS/I/2025/08
	The Venn Diagram below shows three sets, P, Q and R and an element x.  Their relationship can be described using one or more of the following set notations: $\cup \quad \cap \quad \subset \quad \in$ Write down a statement, using the above set notations only, to relate (a) P, Q and R, [1] (b) P, Q and x. [1]
2	SCGS/2025/I/16
	$\mathcal{E} = \{x : x \text{ is a positive integer and } x < 10\}$. The Venn diagram shows the universal set and the elements in some of its subsets.  (a) Describe, in words, the set A. [2] (b) Fill in an appropriate set notation to complete the statement. Answer $\{1, 3\}$ A [1] (c) Find the value of $n((A \cup B)')$. [1]

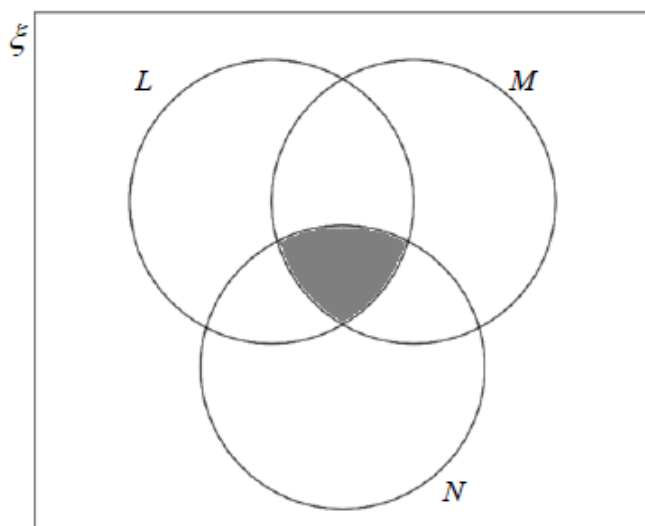
$$\xi = \{1, 2, 3, 4, 8, 15, 27, 43, 49, 121\}$$

$$L = \{\text{Prime Numbers}\}$$

$$M = \{\text{Odd Numbers}\}$$

$$N = \{\text{Square Numbers}\}$$

- (a) Complete the Venn diagram to illustrate this information.
There are no elements in the shaded portion of the diagram.



- (b) List the elements in $(M \cup L)'$
- (c) Find the value of $n((L' \cup N) \cap (L \cup N'))$.
- (d) The numbers in ξ are each printed on one card.
If a card is picked at random from the ten cards, find the
printed on the card
- (i) is a perfect cube,
- (ii) is a multiple of 3.

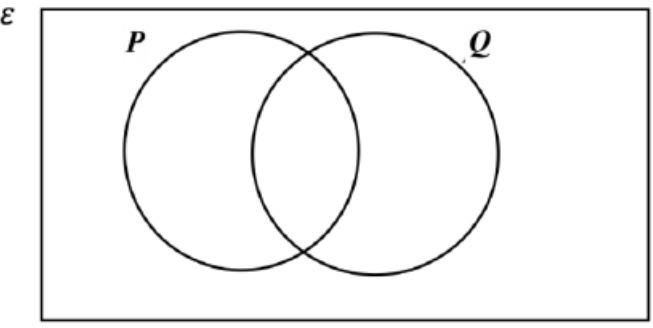
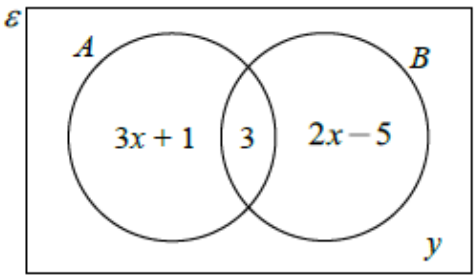
[2]

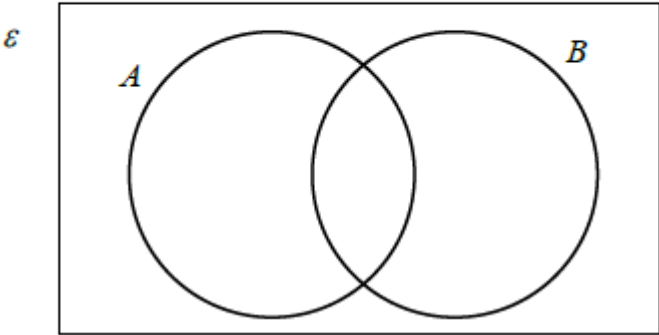
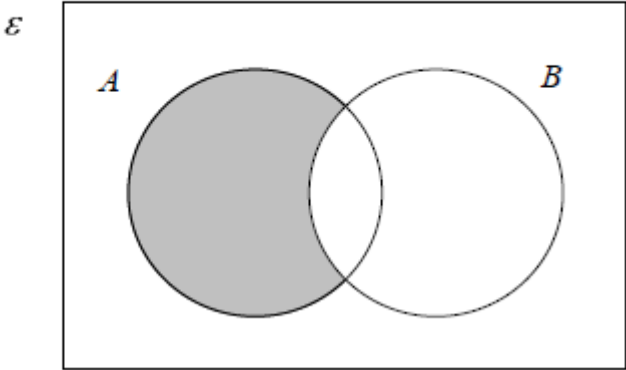
[1]

[1]

[1]

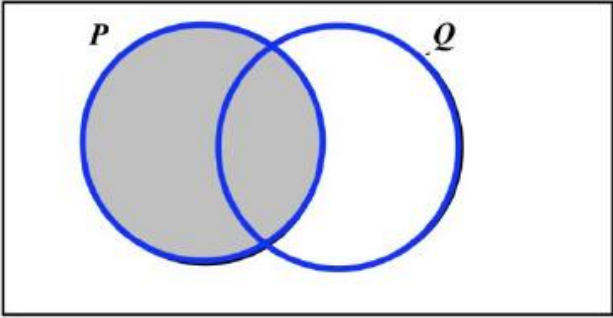
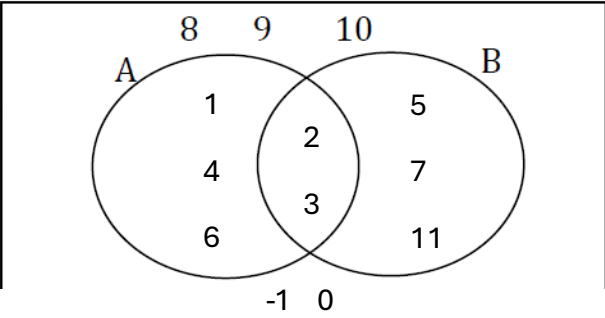
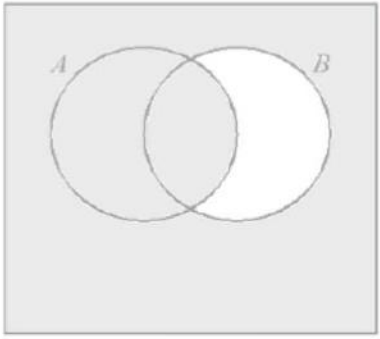
[1]

4	SST/2025/I/01
	$\xi = \{\text{integers } x : 3 < x < 18\}$ $P = \{\text{multiples of } 5\}$ $Q = \{\text{perfect squares}\}$ (a) List the elements in $(P \cup Q)'$. (b) On the Venn diagram, shade the region which represents $(P \cap Q) \cup (P \cap Q')$. [1] <i>Answer</i>  [1]
5	GMSS/2025/I/15
	The Venn diagram shows the universal set and the number of elements in each of the subsets.  Given that $n(A \cup B) = 89$ and $n(E) = 100$, find (a) the value of x and of y, (b) $n(A \cup B' \cap A' \cup B)$. [3] [1]

6	ACSB/2025/I/23 $\xi = \{\text{integers } x : -2 < x \leq 11\}$. $A = \{\text{factors of } 12\}$. $B = \{\text{prime numbers}\}$. (a) Draw a Venn Diagram to represent all the sets above with their elements clearly shown. [2] (b) $n(A \cup B') \cap A' \cup B$. [1]
7	KRSS/2025/I/17 (a) On the Venn Diagram shown in the answer space, shade the set $A \cup B'$. [1] <i>Answer</i>  (b) Write down the set represented by the following shaded region. [1]  (c) [1] $\xi = \{x: x \text{ is an integer and } x \geq 1\}$ $P = \{x: x \text{ is a prime number}\}$ $Q = \{x: x \text{ is a perfect square}\}$ $R = \{x: x \text{ is an integer ending in } 2\}$ (i) Find $n(P \cap R)$. [1] (ii) Find $P \cap Q$. [1]

Solution

1	(a) $R \subset P \cup Q$ (b) $x \in P \cap Q$						
2	<table><tr><td>(a)</td><td>The set of positive odd integers less than 10.</td></tr><tr><td>(b)</td><td>$\{1,3\} \dots \subset \dots A$ ($\subseteq$ not acceptable)</td></tr><tr><td>(c)</td><td>$n((A \cup B)') = 3$</td></tr></table>	(a)	The set of positive odd integers less than 10.	(b)	$\{1,3\} \dots \subset \dots A$ ($\subseteq$ not acceptable)	(c)	$n((A \cup B)') = 3$
(a)	The set of positive odd integers less than 10.						
(b)	$\{1,3\} \dots \subset \dots A$ ($\subseteq$ not acceptable)						
(c)	$n((A \cup B)') = 3$						
3	(a) ξ (b) $\{4, 8\}$ (c) $(L' \cup N) = \{1, 4, 8, 15, 27, 49, 121\}$ $(L \cup N') = \{2, 3, 8, 15, 27, 43\}$ $n((L' \cup N) \cap (L \cup N')) = 3$ (di) (1, 8, 27). Thus the probability is $\frac{3}{10}$. (dii) (3, 15, 27). Thus the probability is $\frac{3}{10}$.						
4	(a) $P \cup Q = \{4, 5, 9, 10, 15, 16\}$ $(P \cup Q)' = \{6, 7, 8, 11, 12, 13, 14, 17\}$ (b)						

	ε	
5	(a) $3x + 1 + 3 + 2x - 5 = 89$ $5x = 90$ $x = 18$ $y + 89 = 100$ $y = 11$ (b) $3 + 11 = 14$	
6	ξ	 $\begin{array}{ccc} 1 & 2 & 5 \\ 4 & 3 & 7 \\ 6 & & 11 \\ & -1 & 0 \end{array}$ (b) $(A \cup B)' = \{-1, 0, 8, 9, 10\}$ (c) $n((A \cup B) \cap (A \cap B)') = 6$
7	a	
	b	$A \cap B'$
	ci	1
	cii	$\emptyset$ or $\{\}$